

Growth Without Spillovers: Metro Size and Uneven Economic Outcomes in North Carolina

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Counties in North Carolina differ considerably in terms of size, economic structure, and access to nearby urban centers. While the characterization of counties as urban or rural is helpful, it often masks a lot of variation. To better capture these differences across counties, this brief uses the Rural-Urban Continuum Codes (RUCC) developed by the U.S. Department of Agriculture. RUCC classifies counties based on three key factors:

- (1) whether a county is part of a metropolitan area,
- (2) the size of the county's urban population and
- (3) whether a non-metropolitan county is adjacent to a metropolitan area.

Together, these factors create a nine-category system that puts counties along a continuum from large metropolitan areas to the most remote rural areas.

In this brief, 1–3 represent metropolitan counties, differentiated by the size of the metro area. RUCC codes 4–9 represent non-metropolitan counties, further distinguished by the size of the urban population and whether they are adjacent to a metro area.

Throughout this brief, we use these RUCC classifications to examine differences in employment growth, wage growth, and industry specialization across North Carolina counties from 2000 to 2024. By doing so, the analysis highlights how geography and proximity to urban

centers shape economic opportunities across the state. Table 1 summarizes the Rural-Urban Continuum Codes

Table 1: Rural-Urban Continuum Codes (RUCC)

RUCC	Type	Urban Population
1	Metro	1M+ population
2	Metro	250k–1M population
3	Metro	< 250k population
4	Non-metro	20k+ urban population, adjacent to metro
5	Non-metro	20k+ urban population, not adjacent
6	Non-metro	5k–20k urban population, adjacent
7	Non-metro	5k–20k urban population, not adjacent
8	Non-metro	< 5k urban population, adjacent
9	Non-metro	< 5k population, not adjacent

(RUCC) used in this brief to classify North Carolina counties by metro status, urban population size, and adjacency to metropolitan areas.

Where Are RUCC Counties Located in North Carolina?

Figure 1 maps North Carolina's 100 counties by RUCC classification. We create four maps: large metros (code 1), small/medium metros (codes 2 and 3), non-metropolitan counties adjacent to metro areas (codes 4, 6, 8), and remote non-metropolitan counties (codes 5, 7, 9). These patterns reveal how population density and proximity to urban centers structure regional economic conditions across the state.

Metropolitan counties cluster around the state's largest urban centers. Mecklenburg and Wake anchor North Carolina's largest labor markets, concentrating jobs, higher wages, and major industries that attract workers from across the state. Even smaller metro counties like Cabarrus and Johnston benefit from their loca-

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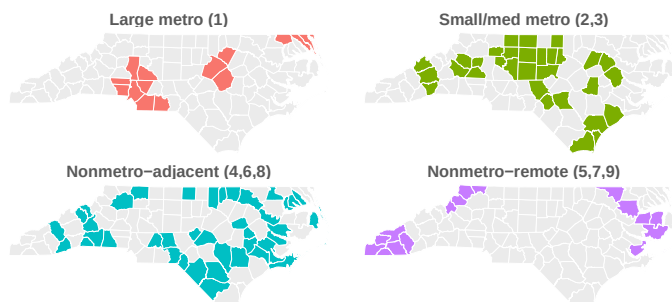


Figure 1: Distribution of North Carolina counties by Rural–Urban Continuum Code (RUCC) group

tion within these broader metropolitan systems.

Surrounding these urban cores are non-metro counties adjacent to metro areas. Counties like Harnett, Cleveland, and Duplin are not metropolitan themselves but lie within commuting distance of a metropolitan area. Many residents travel to nearby metro areas for work, and local businesses often depend on metropolitan markets. These counties frequently experience economic spillovers from nearby urban growth.

Non-metro counties not adjacent to metro areas are more geographically isolated. Hyde, Tyrrell, and Graham exemplify this group, located farther from major employment centers and commuting corridors in eastern North Carolina and the western mountains.

The maps highlight that RUCC captures more than whether a county is urban or rural—it also reflects connectivity to nearby economic centers, an important factor in understanding differences in employment growth, wages, and industry specialization across North Carolina.

Job Growth Patterns Across Metro and Non-Metro Counties

Employment growth across North Carolina has varied dramatically by county type over the past two decades, and the data reveal three patterns worth highlighting (Figure 2).

The Great Recession hit non-metro counties hardest—and unevenly. Between 2000 and 2010, only two county types posted positive employment growth: large metro

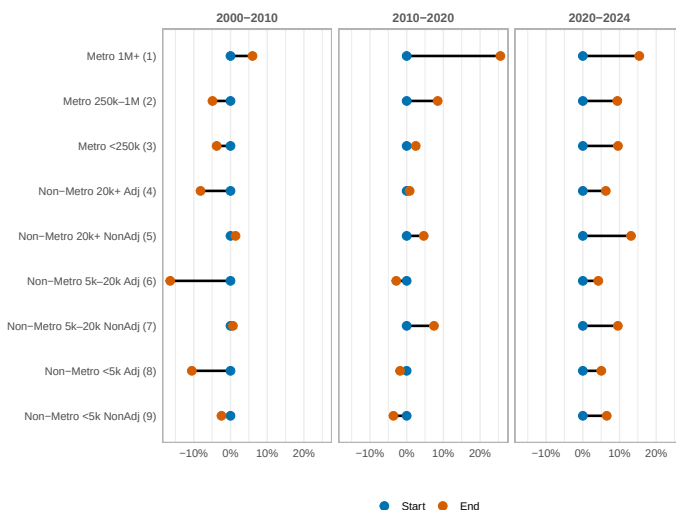


Figure 2: Employment Growth Rate by RUCC relative to 2000

counties (Metro 1M+, +6.0%) and non-metro non-adjacent counties with urban populations between 5,000 and 20,000 (+0.7%) and 20,000 or more (+1.4%). Every other county type lost jobs. The steepest losses fell on metro-adjacent non-metro counties: those with urban populations between 5,000 and 20,000 declined by 16.4%, and smaller adjacent rural counties fell by 10.5%. Adjacency to a metro area, in other words, offered no protection during the downturn—it may have amplified it.

The 2010–2020 recovery was a metro story, with a notable non-adjacent exception. Large metros led decisively, with counties above one million in population growing by 25.5%—more than three times the rate of mid-sized metros (8.5%) and ten times that of smaller metros (2.5%). Among non-metro counties, the adjacency pattern reversed in a striking way: non-adjacent counties consistently outperformed their adjacent counterparts. Non-adjacent counties with urban populations between 5,000 and 20,000 grew by 7.4%, while their adjacent peers actually contracted by 2.8%. The same pattern held at smaller population sizes. This suggests that proximity to a metro area may suppress rather than stimulate local job growth—likely because workers in adjacent counties commute outward, reducing the incentive for local employers to expand.

Since 2020, the geography of growth has broadened. All nine county types posted positive employment growth between 2020 and 2024. Large metros continued to lead (+15.4%), but the gap narrowed considerably. Non-adjacent non-metro counties with populations above 20,000 grew by 13.1%—nearly on par with the largest metros—while mid-sized metros (+9.4%) and smaller metros (+9.6%) posted nearly identical gains.

Even the weakest performers, metro-adjacent counties with urban populations between 5,000 and 20,000, grew by 4.2%. Whether this broadening reflects a structural shift—remote work, housing costs pushing residents outward—or a cyclical post-pandemic rebound remains an open question.

Wage Growth Patterns: Convergence After Decades of Divergence

Wage growth across North Carolina’s counties reveals a striking shift in recent years. Figure 4 compares wage growth rates across county types for three periods: 2000–2010, 2010–2020, and 2020–2024.

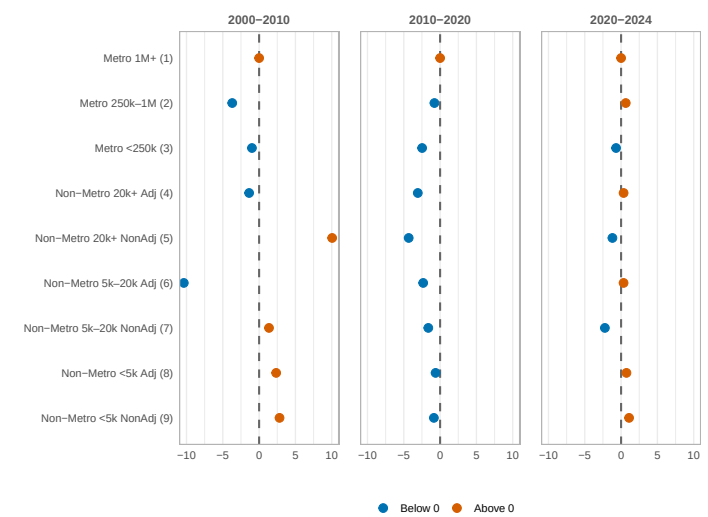


Figure 3: Wage growth rates by county type and time period. Each bar shows the percentage change in average weekly wages for that county type during the specified period.

The 2000–2010 period showed surprisingly strong non-metro wage performance. Large metros posted 30.5% wage growth, but several non-metro county types exceeded this benchmark. Non-adjacent counties with populations above 20,000 grew wages by 40.5% — 10.1 percentage points *above* large metros. The smallest non-adjacent rural counties also outperformed, growing 33.3% (+2.8 pp above large metros). In contrast, metro-adjacent non-metro counties with populations between 5,000 and 20,000 lagged dramatically, posting only 20.1% growth — 10.4 percentage points below the large metro benchmark. Interestingly, geographic isolation from metros correlated with stronger wage growth during this decade, while adjacency provided no advantage.

The 2010–2020 recovery inverted this pattern entirely. Large metros led with 31.8% wage growth, and this time every other county type fell short. The gaps were mod-

est but consistent: mid-sized metros lagged by 0.8 percentage points, smaller metros by 2.5 pp, and non-metro counties by 1.6 to 4.3 percentage points. Non-adjacent counties with populations above 20,000 — the strongest performers in 2000–2010 — now posted the weakest wage growth among non-metro counties, falling 4.3 pp behind large metros. The metro advantage reasserted itself across the board.

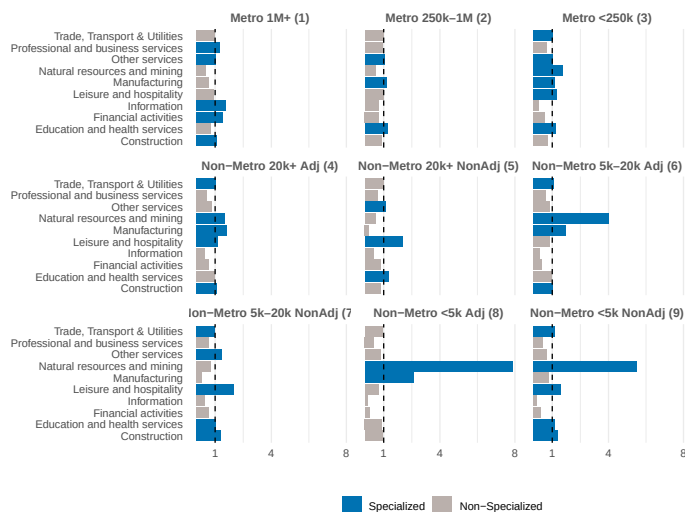
Since 2020, wage growth differences have nearly vanished. Large metros grew wages by 20.6%, but the spread across all nine county types compressed to just 3.3 percentage points — the tightest clustering in the entire 24-year period. Mid-sized metros exceeded large metros by 0.6 pp. The smallest non-adjacent rural counties grew wages by 21.7% (+1.1 pp). Even the weakest performer, small non-adjacent non-metro counties at 18.3%, lagged large metros by only 2.2 pp, far smaller than the 4–10 percentage point gaps common in prior decades.

This convergence is remarkable not for eliminating wage level differences as large metros still pay substantially higher absolute wages but for compressing growth rate differences to near-zero. Whatever mechanisms drove the metro wage premium during the 2010s appear to have weakened considerably since 2020.

What Industries Drive the metro and non-metro counties?

The employment growth patterns discussed above are closely tied to what kinds of industries dominate local economies. Figure 4 shifts the focus from growth rates to industry specialization, showing which sectors account for a larger share of employment in different county types relative to the state as a whole. In metropolitan counties, industry specialization is relatively balanced. Large and mid-sized metro areas show concentrations in professional and business services, information, and education and health services, reflecting more diversified labor markets. No single industry overwhelmingly dominates employment in these counties, which helps explain why metropolitan areas were better positioned to absorb job losses and recover more quickly during economic downturns.

Non-metro counties, particularly those farther from metropolitan areas, display much sharper specialization patterns. Several non-metro groups show strong concentration in natural resources and mining, with some industries accounting for a disproportionately large share of local employment. This pattern is most visible in small, remote counties such as Avery, where mining-related activities dominate the local job base. In these counties, a single sector often supports a large share of



Taken together, the evidence suggests that metro-led expansion is not sufficient to generate balanced statewide growth. Economic outcomes are shaped not only by location, but by diversification and local capacity for job creation. Understanding these patterns is essential for interpreting regional disparities and setting realistic expectations about how growth distributes across North Carolina.

Figure 4: Industry location quotients (LQ) for major sectors across North Carolina counties grouped by Rural–Urban Continuum Code (RUCC)

employment, leaving local economies highly exposed to downturns in that industry. This concentration increases vulnerability to sector-specific shocks and helps explain why many non-metro counties experienced weaker and less consistent employment growth over time. These specialization patterns highlight an important tradeoff. Counties with more diversified industry bases, mainly metropolitan areas, tend to experience more stable employment growth, whereas counties with narrower economic structures are more exposed to downturn-related risks.

Why does any of this matter?

The central finding of this analysis is that economic growth in North Carolina is increasingly concentrated in its largest metropolitan counties, and that growth does not automatically spill over into surrounding areas. Several metro-adjacent counties underperformed even some remote rural counties during recovery periods, challenging the assumption that proximity to a major city guarantees local job creation. Geography matters, but connectivity alone does not ensure shared growth.

Industry structure helps explain these differences. Metropolitan counties benefit from more diversified economies, which has supported stronger and more stable employment growth over time. Many non-metro counties remain concentrated in a narrow set of industries, leaving them more vulnerable to downturns and limiting long-term resilience. While recent wage growth in some rural counties has narrowed gaps with large metros, differences in employment scale and economic composition persist.